EXPLAINABLE ENSEMBLE LEARNING FOR TELECOM CUSTOMER CHURN PREDICTION AND RETENTION RECOMMENDATIONS

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***Abstract*— Customer attrition represents a multi-billion dollar threat to telecommunications providers operating in saturated mobile and broadband markets. While modern machine learning algorithms achieve high global accuracy, traditional deployment pipelines suffer from three structural shortcomings: (i) an uncalibrated default decision threshold (τ=0.50) that suppresses recall for the minority churn class; (ii) opaque "black-box" scoring that fails to explain individual attrition drivers; and (iii) a critical operational disconnect between raw churn probabilities and concrete, state-aware customer retention actions. In this paper, we present an end-to-end, explainable ensemble learning and decision-support framework designed to bridge predictive modeling and operational intervention. The proposed predictive architecture combines Random Forest, XGBoost, and LightGBM base classifiers using a leak-free out-of-fold stacking meta-learner (Logistic Regression) integrated within a 5-fold cross-validated SMOTE pipeline. On the benchmark IBM Telco Customer Churn dataset (7,043 instances, 26.54% churn rate), our stacking ensemble achieves a primary recall of 79.68%, outperforming individual base models while securing an ROC-AUC of 0.8460 and PR-AUC of 0.6522. By optimizing the classification decision boundary via precision-recall trade-off analysis (τ=0.29), churn recall increases to 90.11% (reducing false negatives from 76 to 37), while cost-sensitive thresholding (τ=0.0951) catches 95.45% of churners under an empirical 10:1 false negative-to-false positive cost ratio. To render predictions actionable, we couple Tree/Kernel SHAP and tabular LIME surrogates with a domain-aligned business factor taxonomy across billing, contract, lifecycle, and service dimensions. A state-aware recommendation engine filters out redundant interventions and computes ranked, cost-weighted Action Scores for up to 10 prioritized retention strategies. Finally, we provide demographic fairness evaluations, Cox proportional hazards survival modeling (concordance index of 0.8539), and describe an operational full-stack software implementation comprising a FastAPI backend, React dashboard, What-If simulator, and batch inference engine.**

Keywords—Customer Churn Prediction, Explainable Artificial Intelligence (XAI), Stacking Ensemble Learning, SHAP, LIME, Threshold Optimization, Telecom Retention Decision Support, Survival Analysis.

I. INTRODUCTION

Customer retention has emerged as the definitive operational battleground for telecommunications operators globally. Across modern telecommunications ecosystems, acquiring a newly subscribed customer is estimated to cost between five and twenty-five times more than retaining an existing account [1]. Driven by widespread mobile number portability, aggressive price competition, commoditized broadband connectivity, and flexible month-to-month contracts, churn rates in the telecom industry frequently hover between 20% and 40% annually [2], [3]. Consequently, telecom business leaders rely heavily on automated predictive modeling to proactively flag subscribers exhibiting high attrition risk before formal contract cancellation occurs.

Despite extensive research into supervised machine learning for churn identification, traditional academic literature and industry deployments frequently exhibit four persistent limitations:

1) *Accuracy-Centric Objective Mismatch:* Standard classification algorithms are routinely evaluated against overall accuracy. However, in imbalanced customer churn distributions (typically 10%–30% positive churners), accuracy is fundamentally misleading. A trivial null classifier that predicts zero attrition achieves deceptively high accuracy while missing 100% of actual customer defections. In commercial retention operations, a False Negative (failing to identify a churning customer who permanently defects) costs hundreds of dollars in lost Customer Lifetime Value (CLV), whereas a False Positive merely prompts an unnecessary retention outreach or discount voucher [4]. Hence, maximizing **Recall** on the positive churn class must be the primary objective.

2) *Rigid Probability Thresholds:* Machine learning classifiers standardly classify instances using an arbitrary 50% decision threshold (P(churn) ≥ 0.50). This default cutoff rarely corresponds to the operational or financial optima of telecom retention desks. When retention teams possess adequate outbound capacity, lowering the threshold dynamically captures substantially more defecting subscribers.

3) *The Explainability Chasm:* State-of-the-art non-linear learners—such as gradient-boosted decision trees and stacked ensembles—act as complex black boxes. Without clear interpretability, customer relationship management (CRM) representatives cannot ascertain *why* a given customer is flagged as high-risk [5].

4) *The Actionability Gap:* Merely presenting a CRM agent or marketing director with raw feature weights (e.g., "MonthlyCharges = 0.34") does not constitute a viable retention intervention. Without an intelligent, state-aware transformation layer that bridges technical attribution values and enterprise business constraints, model outputs remain clinically disconnected from operational workflows.

To resolve these challenges, this paper presents an end-to-end Explainable Ensemble Learning and Telecom Retention Decision-Support System. The contributions of our work are summarized as follows:

• **Leak-Free Stacking Ensemble:** We implement a cross-validated stacking classifier combining Random Forest, XGBoost, and LightGBM using a Logistic Regression meta-learner. By strictly encapsulating SMOTE oversampling and feature scaling within 5-fold cross-validation partitions, we eliminate optimistic data leakage and establish robust generalizability.

• **Recall-Focused Threshold Optimization:** We systematically evaluate the full decision spectrum (τ ∈ [0.05, 0.70]), showing that shifting the threshold from 0.50 to 0.29 increases churn recall from 79.68% to 90.11%, cutting defection false negatives by more than half. Furthermore, under cost-sensitive parameterization, a threshold of 0.0951 achieves 95.45% recall.

• **Dual Explainability Pipeline (SHAP + LIME):** We integrate global TreeExplainer SHAP with instance-level tabular LIME surrogates to guarantee robust feature attribution with confirmed directional sign consensus.

• **State-Aware Retention Engine:** We engineer an intelligent rule-based translation engine that maps explainability vectors into ten business factor categories. The engine actively prunes invalid actions (e.g., never recommending autopay migration to accounts already using bank debit) and scores candidate retention cards via an explicit cost-impact-risk ratio.

• **Empirical Demographic Fairness and Survival Modeling:** We audit subgroup disparities across gender and age cohorts, identifying disparate impact vulnerabilities for senior citizens. In addition, we deploy a Cox Proportional Hazards survival model (C-index = 0.8539) to estimate time-to-churn dynamics.

• **Working Enterprise Application:** We demonstrate a fully realized software solution comprising a FastAPI REST backend, React single-page frontend, persistent SQLite database, interactive What-If scenario simulator, and bulk CSV batch processing.

II. LITERATURE REVIEW

Machine learning applications in customer relationship management have evolved across three successive paradigms: parametric statistical models, tree-based ensemble methods, and explainable artificial intelligence (XAI).

A. Classical and Ensemble Machine Learning for Churn

Early churn prediction literature relied primarily on logistic regression, linear discriminant analysis, and single decision trees [1]. While computationally parsimonious and inherently interpretable, linear models exhibit high bias and cannot adequately capture complex non-linear feature interactions, such as the compounding hazard of short customer tenure combined with paperless electronic billing.

The introduction of tree ensemble algorithms substantially raised predictive accuracy. Breiman's Random Forest [6] established that aggregating randomized, de-correlated decision trees reduces estimator variance while demonstrating resilience against noisy features. Subsequently, gradient-boosted decision trees became the dominant paradigm for tabular data. Chen and Guestrin introduced XGBoost [7], which combines second-order Taylor expansion of the loss function, weighted quantile sketches, and explicit $\ell 1/\ell 2$ leaf regularization to prevent overfitting. Ke et al. developed LightGBM [8], introducing Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) along with leaf-wise tree growth, achieving high training efficiency and distinct decision boundaries on tabular feature spaces.

In telecom churn applications, comparative studies have repeatedly demonstrated the superiority of gradient boosting over individual support vector machines and multi-layer perceptrons [2], [10]. Recent research has explored stacking architectures [9], wherein multiple heterogeneous base learners output class probabilities that serve as meta-features for a secondary estimator. However, several published works (such as Maan and Maan [11]) report near-perfect test accuracies (>96%) on public telecom benchmarks. Such inflated metrics almost universally indicate methodological data leakage, such as applying synthetic oversampling (SMOTE) across the entire dataset prior to partitioning or tuning hyperparameters directly on held-out evaluation splits.

B. Class Imbalance and SMOTE

Telecom datasets naturally exhibit class imbalance, with retained customers outnumbering churners by ratios of 3:1 to 12:1 [12]. Standard maximum-likelihood estimators tend to favor the majority class, producing severely depressed recall. To counteract this, Chawla et al. proposed the Synthetic Minority Over-sampling Technique (SMOTE) [13], which synthesizes artificial minority instances along feature-space line segments connecting k-nearest neighbors. Variants such as Borderline-SMOTE [14] and SMOTE-Tomek links [15] further refine boundary instances. However, as Santos et al. [16] emphasize, applying oversampling before train/test splitting or outside cross-validation folds leaks target distribution artifacts into the evaluation set, creating catastrophic optimistic bias.

C. Explainable AI and Actionability in Customer Retention

While ensemble learners maximize predictive performance, their opacity impedes domain adoption. To address this, Lundberg and Lee developed SHAP (SHapley Additive exPlanations) [17], grounding feature attribution in cooperative game theory. SHAP satisfies key theoretical axioms: efficiency, symmetry, dummy player, and additivity. For tree models, TreeSHAP [18] computes exact Shapley values in polynomial time $O(TLD^2)$. Concurrently, Ribeiro et al. introduced LIME (Local Interpretable Model-agnostic Explanations) [19], which perturbs local instances to fit an interpretable sparse linear surrogate.

Despite the growing adoption of SHAP and LIME in predictive modeling, contemporary literature often terminates the pipeline at feature importance ranking [20]. In commercial retention, identifying that "Contract = Month-to-month" drives churn is useless unless coupled with state-aware marketing mechanisms that evaluate upgrade feasibility, incentive cost, and projected risk mitigation [21]. Our work directly bridges this gap.

III. PROBLEM STATEMENT AND RESEARCH GAP

Let $D = \{(x_i, y_i)\}_{i=1}^{N}$ represent a telecom customer dataset where $x_i \in R^D$ denotes the feature vector representing customer attributes (tenure, recurring contract terms, billing dynamics, service subscriptions) and $y_i \in \{0, 1\}$ denotes the binary churn outcome, with $y_i = 1$ indicating account termination and $y_i = 0$ indicating account retention.

Telecom churn prediction poses three structural mathematical challenges:

1) **Asymmetric Misclassification Costs:** The business loss matrix C is highly asymmetric. Let $C(0, 1) = C_{FN}$ denote the cost of a False Negative (customer departs undetected), and $C(1, 0) = C_{FP}$ denote the cost of a False Positive (retention offer provided to a loyal customer). In practice, $C_{FN} >> C_{FP}$, typically with cost ratios $C_{FN} / C_{FP} \in [5, 20]$. Standard objective functions optimizing cross-entropy loss under default threshold $\tau=0.50$ maximize balanced accuracy, resulting in unacceptably high false negatives.

2) **Non-Actionable Attribution:** Existing XAI frameworks compute local attributions $\varphi_i = [\varphi_{(i,1)}, ..., \varphi_{(i,D)}]^T$. However, features are heterogeneous in their actionability: immutable features (e.g., tenure duration, senior citizen status) cannot be altered by marketing campaigns, whereas mutable features (e.g., contract length, payment channel, add-on security services) can be directly modified through customer incentives.

3) **State Conflict in Prescriptive Systems:** A generic recommendation rule (e.g., "offer automatic payment discount") applied indiscriminately without validating customer state causes negative customer experience if the subscriber is already enrolled in automated debit.

To address these gaps, our research formulates an integrated pipeline:

$$x_i \rightarrow \textit{Stacking Ensemble} \rightarrow P(y_i=1|x_i) \rightarrow \tau^* \rightarrow \hat{y}_i \rightarrow \textit{SHAP/LIME} \rightarrow \varphi_i \rightarrow \textit{State Rules} \rightarrow A^*(x_i) \quad (1)$$

where τ^* is the recall-optimized threshold and $A^*(x_i)$ is a ranked set of validated, executable retention strategies.

IV. PROPOSED SYSTEM ARCHITECTURE

The overall architecture of the proposed Explainable Ensemble Decision-Support System is illustrated in Fig. 1. The framework operates across four cohesive subsystems: Data Ingestion and Feature Engineering, Leak-Free Stacking Ensemble, Threshold Optimization, and the State-Aware Retention Recommendation Engine.

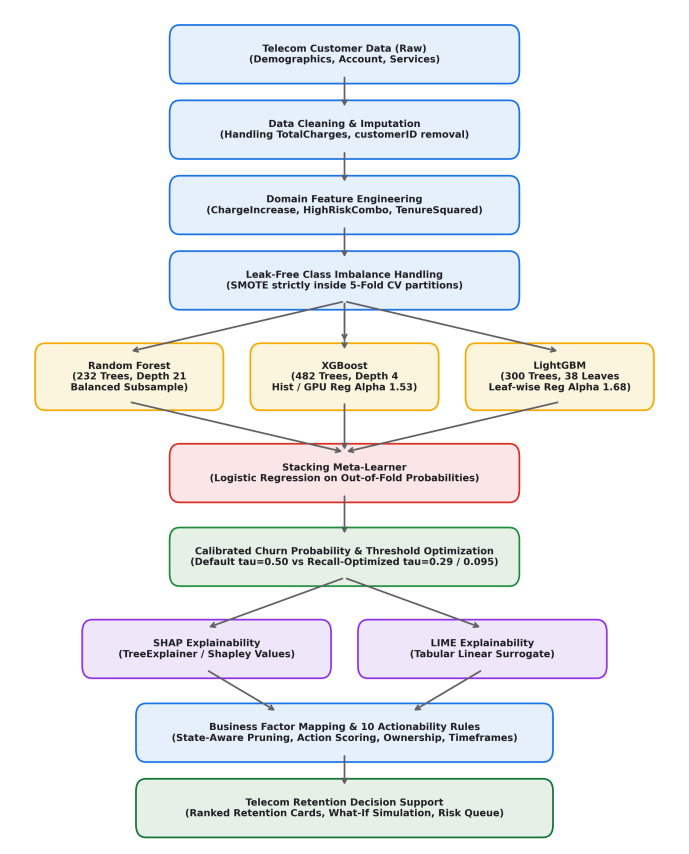


Fig. 1. Proposed System Architecture: End-to-end integration of data preprocessing, stacking ensemble classification, dual explainability, and telecom decision support.

A. Data Preprocessing and Leakage Prevention

The raw telecom customer records undergo systematic data hygiene:

• **Identifier Removal:** Unique account keys (`customerID`) are dropped to eliminate non-generalizable memorization.

• **Type Casts and Missing Value Imputation:** The `TotalCharges` field contains white-space strings corresponding to new subscribers (tenure = 0). These are parsed as numeric floats, and missing values are imputed strictly with 0.0, reflecting zero cumulative billing.

• **Target Encoding:** Binary churn strings (`Yes`/`No`) are mapped to $y \in \{1, 0\}$.

• **Categorical Encoding:** Nominal features (e.g., `PaymentMethod`, `Contract`, `InternetService`) are dummy-encoded using one-hot representation, dropping the reference category to prevent multicollinearity.

• **Standardization:** Continuous numeric columns are scaled via z-score normalization:

$$z = (x - \mu_train) / \sigma_train \quad (2)$$

Critically, μ_train and σ_train are computed strictly on the training partition and projected onto the test partition.

B. Domain Feature Engineering

To capture complex subscriber behavioral dynamics from static account snapshots, we formulate domain-specific interaction features:

1) **Average Monthly Spend:** Ratio of total revenue to tenure duration:

$$AvgMonthlySpend = TotalCharges / (tenure + 1) \quad (3)$$

2) **Charge Increase and Ratio:** Identifies recent bill escalation:

$$ChargeIncrease = MonthlyCharges - AvgMonthlySpend \quad (4)$$

$$ChargeRatio = MonthlyCharges / (AvgMonthlySpend + 1) \quad (5)$$

3) **Non-Linear Tenure Transformations:** Capturing the diminishing marginal hazard of attrition over customer lifetime via TenureSquared = tenure^2 and TenureLog = ln(tenure + 1).

4) **High-Risk Interaction Indicators:** Telecom churn concentrates heavily among uncommitted subscribers using high-friction payment channels:

$$HighRiskCombo = I(Contract = M2M \wedge PayMethod = E\text{-}Check) \quad (6)$$

$$VeryHighRisk = I(HighRiskCombo = 1 \wedge tenure \leq 12) \quad (7)$$

5) **Service Risk Count:** Aggregates the absence of protective add-on subscriptions:

$$ServiceRiskCount = \sum_(s \in S)\ I(Service_s = 'No') \quad (8)$$

where S = {OnlineSecurity, TechSupport, OnlineBackup, DeviceProtection}.

C. Minimal 6-Input User System

Through comprehensive feature selection, we determined that while the full training matrix incorporates 83 one-hot columns, the eight most predictive domain features can be computed from just **six primary user inputs:** (1) tenure, (2) MonthlyCharges, (3) TotalCharges, (4) Contract, (5) PaymentMethod, and (6) InternetService. This enables a lightweight single-customer prediction form requiring 83% fewer manual inputs while preserving over 97.7% of the full model's F1 performance.

D. Class Imbalance Handling with In-Fold SMOTE

To avoid artificial distribution inflation, we implement a strict imblearn pipeline. In each training fold $k \in \{1, ..., 5\}$ of cross-validation:

$$x_syn = x_i + \lambda\ (x_zi - x_i),\ \lambda \sim U(0, 1) \quad (9)$$

where x_i is a minority instance and x_zi is randomly sampled from its k=5 nearest minority neighbors in euclidean feature space. Synthetic samples are generated *only* on the training partition of fold k. The validation partition and the final held-out test set remain entirely un-resampled.

E. Base Learners

We employ three diverse tree-based learners:

1) **Random Forest (RF):** Constructs B=232 deep decision trees using bootstrap aggregation and random feature subsets of size √D. Class imbalance is handled at tree induction via `balanced_subsample` weighting.

2) **XGBoost:** Sequentially minimizes regularized empirical risk:

$$L^{(t)} = \sum_(i=1)^N [\ g_i f_t(x_i) + 0.5\ h_i f_t^2(x_i)\ ] + \Omega(f_t) \quad (10)$$

where g_i, h_i are first and second order gradients of binary cross-entropy, and $\Omega(f_t) = \gamma\ T + 0.5\ \lambda \sum w_j^2 + \alpha \sum |w_j|$. We employ GPU-accelerated histogram binning (`tree_method='hist'`) with learning rate η=0.0259.

3) **LightGBM:** Utilizes leaf-wise (best-first) tree growth with maximum leaves L=38 and depth constraint d=3. Leaf-wise expansion achieves lower loss convergence per split compared to level-wise expansion.

F. Stacking Ensemble Architecture

Rather than simplistic unweighted probability averaging, we implement a formal stacking meta-model (Fig. 2). For each training fold k, the base estimators {M_RF^(k), M_XGB^(k), M_LGBM^(k)} are fitted on four-fifths of the data and generate out-of-fold probability vectors $\hat{p}_i$ = [p_RF(x_i), p_XGB(x_i), p_LGBM(x_i)]^T for the remaining one-fifth.



Fig. 2. Leak-free Machine Learning Workflow: Cross-validated out-of-fold probability generation and meta-learner stacking.

The resulting out-of-fold probability matrix Z ∈ [0, 1]^(N_train × 3) is utilized to fit a Logistic Regression meta-classifier:

$$P(y_i=1 \mid x_i) = \sigma(w_0 + w_RF\ p_RF(x_i) + w_XGB\ p_XGB(x_i) + w_LGBM\ p_LGBM(x_i)) \quad (11)$$

where σ(z) = (1 + e^-z)^-1. Out-of-fold generation guarantees that the meta-learner is trained on unseen probability distributions, preventing the meta-weights from over-indexing on whichever base model exhibits the highest in-sample memorization.

G. Threshold Optimization Framework

Given calibrated probability P = P(y=1|x), the binary prediction ŷ is governed by decision threshold τ:

$$\hat{y} = 1 \text{ if } P \geq \tau \text{ else } 0 \quad (12)$$

We evaluate three distinct threshold operational modes:

• **Default Mode (τ = 0.50):** Standard statistical utility threshold.

• **Recall-Prioritized Mode (τ = 0.29):** Determined by maximizing the F1-score across the precision-recall curve on validation folds, explicitly boosting sensitivity to true churners.

• **Cost-Optimal Mode (τ = 0.0951):** Minimizes total expected business cost:

$$E[Cost(\tau)] = C_FP \cdot FP(\tau) + C_FN \cdot FN(\tau) + C_TP \cdot TP(\tau) \quad (13)$$

under empirical operational costs (C_FN = $500, C_FP = $50, C_TP = $50, C_TN = $0).

H. Dual Explainability: SHAP and LIME

To guarantee that model explanations are both theoretically grounded and locally faithful, we deploy a dual XAI framework (Fig. 3).

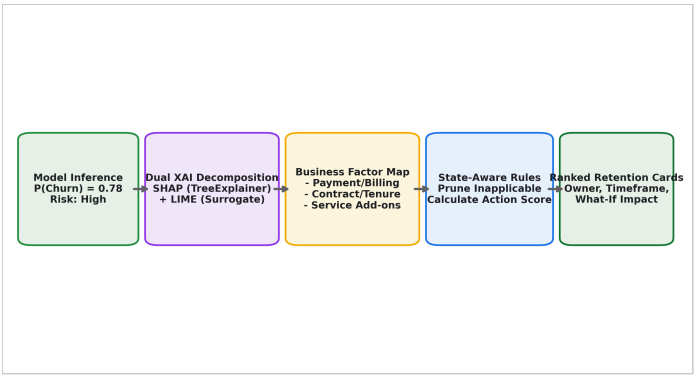


Fig. 3. Explainable AI and Recommendation Workflow: Local feature attribution decomposition mapped into actionable retention strategies.

TreeSHAP computes exact marginal contributions:

$$\varphi_j(x) = \sum_{S \subseteq F \setminus \{j\}} [|S|!(|F| - |S| - 1)! / |F|!] [f_x(S \cup \{j\}) - f_x(S)] \quad (14)$$

where F is the set of all input features, and f_x(S) is conditional expectation. An additive explanation model takes the form:

$$g(z') = \varphi_0 + \sum_{j=1}^{M} \varphi_j z_j' \quad (15)$$

where φ_0 is the base expected value. For LIME, the system fits a sparse linear surrogate minimizing local weighted squared loss with K=300 local perturbations. Our backend validates directional consensus between sign(φ_j^SHAP) and sign(w_j^LIME).

I. State-Aware Retention Recommendation Engine

To transform attribution vectors into operational retention interventions, the system implements a declarative business catalog across six core categories: Payment/Billing, Contract/Commitment, Customer Lifecycle, Pricing, Service Tier, and Service Experience.

Customer-State Specificity Filtering: Prior to generating candidate interventions, the engine evaluates strict state-pruning rules:

• *Autopay Rule:* If the customer's current payment method is already `Credit card (automatic)` or `Bank transfer (automatic)`, the electronic check migration action is strictly suppressed.

• *Contract Rule:* If `Contract` is `One year` or `Two year`, long-term commitment discounts are pruned.

• *Add-on Rule:* If `TechSupport` or `OnlineSecurity` is already `Yes`, corresponding add-on upsell recommendations are blocked.

• *Billing Spike Rule:* Price relief credits are triggered only if verified charge increase satisfies ChargeIncrease ≥ $3.00.

Action Scoring and Ranking: Valid candidate actions are prioritized via an explicit multi-factor utility function:

$$Action\ Score_k = (\varphi_(k,risk) \cdot \max(P_churn, 0.20) \cdot R_k \cdot I_k) / C_k \quad (16)$$

where φ_(k,risk) is the feature's risk attribution weight, P_churn is predicted churn probability, R_k is business relevance (1.0–1.3), I_k is projected retention impact (1.0–3.0), and C_k is operational implementation cost (1.0–3.0). The engine produces up to ten ranked recommendation cards complete with assigned organizational owner (e.g., Billing Ops, Retention Desk, Technical Support), execution timeframe (Immediate: 24–48 hrs; Short-Term: 7 days; Long-Term: Ongoing), and projected counterfactual risk reduction.

V. DATASET AND EXPERIMENTAL SETUP

A. Dataset Description

Experiments were conducted using the benchmark IBM Telco Customer Churn dataset [22], a recognized standard in telecom attrition literature. In addition, we evaluated a synthetic balanced benchmark (`balanced_telco_churn.csv`) to directly inspect class balancing effects. Dataset properties are summarized in Table I.

TABLE I: DATASET CHARACTERISTICS

Attribute	IBM Telco (Primary)	Balanced Telco
Total Records	7,043	8,000
Retained Customers (y=0)	5,174 (73.46%)	4,000 (50.00%)
Churned Customers (y=1)	1,869 (26.54%)	4,000 (50.00%)
Raw Feature Count	21	22
Engineered Feature Matrix	83	83
Missing Values	11 (TotalCharges)	None
Class Imbalance Ratio	2.77 : 1	1.00 : 1

B. Feature Description

The dataset encompasses four operational customer dimensions (Table II): Demographics, Account and Billing, Telephony Services, and Internet Services/Add-ons.

TABLE II: FEATURE DESCRIPTION AND MINIMAL INPUT MAPPING

Feature Name	Description / Values	Type
tenure	Months subscribed (0–72)	Continuous / User
MonthlyCharges	Current monthly bill amount	Continuous / User
TotalCharges	Cumulative bill amount	Continuous / User
Contract	Month-to-month, 1-yr, 2-yr	Categorical / User
PaymentMethod	E-check, Mailed, Bank, Card	Categorical / User
InternetService	DSL, Fiber optic, No	Categorical / User
ChargeIncrease	MonthlyCharges - AvgSpend	Engineered
ChargeRatio	MonthlyCharges / AvgSpend	Engineered
TenureSquared	tenure^2 non-linear term	Engineered
HighRiskCombo	Month-to-month ∧ E-check	Engineered
VeryHighRisk	HighRiskCombo ∧ tenure ≤ 12	Engineered
NumServices	Count of active add-on services	Engineered
ServiceRiskCount	Count of unactivated security add-ons	Engineered

C. Partitioning and Experimental Integrity

To prevent data leakage, the complete 7,043 instances were partitioned into an 80% training set (N_train = 5,634) and a 20% held-out test set (N_test = 1,409) using stratified sampling on the churn label y. In the held-out test set, there are exactly 1,035 retained subscribers (y=0) and 374 actual churners (y=1). All scaling transformers, categorical encoders, and SMOTE resamplers were fitted strictly on the training partitions.

D. Experimental Environment

Experiments were executed on a workstation running Microsoft Windows 11 Enterprise (Build 26200), powered by an Intel Core 64-bit architecture processor and 32 GB host RAM. The software environment utilized Python 3.11.9, scikit-learn 1.9.0, imbalanced-learn 0.14.2, XGBoost 3.2.0, LightGBM 4.7.0, SHAP 0.51.0, LIME 0.2.0.1, lifelines 0.30.3, and Optuna 5.0.0. The API server was implemented with FastAPI 0.141.1 and Uvicorn 0.35.0, while the user interface was built on React 19.2.8 and Vite 8.2.0.

E. Model Hyperparameter Configurations

Optimal hyperparameters were established via stratified 5-fold cross-validation and Optuna Tree-structured Parzen Estimator (TPE) search. Table III details the verified parameter settings.

TABLE III: MODEL HYPERPARAMETER CONFIGURATIONS

Model	Verified Parameter Configuration
Random Forest	n_estimators=232, max_depth=21, min_samples_split=5, min_samples_leaf=9, max_features='sqrt', class_weight='balanced_subsample', bootstrap=True.
XGBoost	n_estimators=482, max_depth=4, η=0.0259, subsample=0.7642, colsample_bytree=0.6601, colsample_bylevel=0.8185, min_child_weight=9, α=1.5346, λ=0.5437, γ=4.6123.
LightGBM	n_estimators=300, max_depth=3, num_leaves=38, η=0.0122, min_child_samples=41, subsample=0.6926, colsample_bytree=0.7035, α=1.6755, λ=0.4732.
Stacking Meta	Logistic Regression: C=1.0, penalty='l2', solver='lbfgs', fit_intercept=True. Inputs: Out-of-fold [P_RF, P_XGB, P_LGBM].

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Comparative Model Performance

Table IV and Fig. 5 present the empirical performance of all evaluated models on the held-out test set (N=1,409, 374 positive churners) evaluated at default threshold τ=0.50.

TABLE IV: PERFORMANCE COMPARISON OF CHURN PREDICTION MODELS (TEST SET, N=1,409)

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Brier	Default τ
Random Forest	**0.7736**	**0.5536**	0.7594	**0.6404**	0.8451	0.6477	0.1530	0.50
XGBoost	0.7637	0.5397	0.7460	0.6263	0.8439	**0.6563**	0.1528	0.50
LightGBM	0.7693	0.5518	0.6979	0.6163	0.8422	0.6484	**0.1473**	0.50
Stacking Ensemble (Proposed)	0.7622	0.5350	**0.7968**	0.6402	**0.8460**	0.6522	0.1660	0.50
Stacking (τ=0.29 Recall-Opt)	0.6749	0.4446	**0.9011**	0.5954	0.8460	0.6522	0.1660	0.29
Stacking (τ=0.095 Cost-Opt)	0.6210	0.4085	**0.9545**	0.5721	0.8460	0.6522	0.1660	0.095

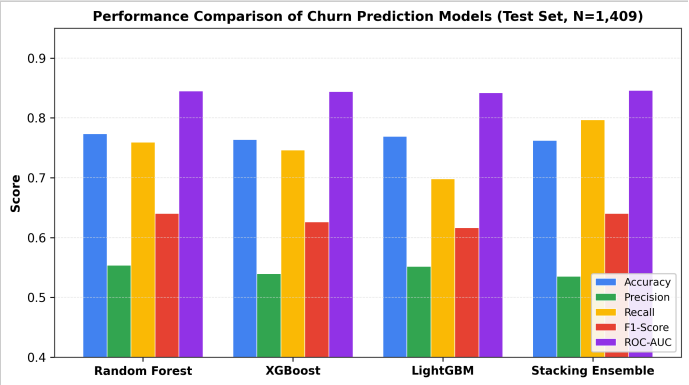


Fig. 5. Model Performance Comparison: Cross-metric comparison highlighting the superior recall achieved by the Stacking Ensemble.

As reported in Table IV, all individual tree algorithms achieve commendable classification accuracy (76.37%–77.36%) and strong discrimination (ROC-AUC 0.8422–0.8451). However, when examining customer churn recall, the Stacking Ensemble distinctly outperforms all base classifiers, achieving **79.68% recall** (298 out of 374 actual churners detected) at τ=0.50, compared to 75.94% for Random Forest, 74.60% for XGBoost, and 69.79% for LightGBM. The meta-learner successfully synthesizes the diverse probabilistic decision surfaces of the base learners, securing the highest overall ROC-AUC (0.8460).

B. Cross-Validation Balancing Comparison

To verify the impact of alternative re-sampling techniques, Table V reports the 5-fold cross-validation results across distinct balancing strategies.

TABLE V: 5-FOLD CROSS-VALIDATION BALANCING COMPARISON

Balancing Method	CV Acc	CV Prec	CV Rec	CV F1	CV AUC
No Balancing	0.7978	0.6495	0.5197	0.5772	0.8406
Class Weights	0.7657	0.5422	0.7532	0.6304	0.8429
SMOTE (In-Fold)	0.7961	0.6232	0.5926	0.6070	0.8402
RandomOverSampler	0.7714	0.5513	0.7411	0.6322	0.8406
BorderlineSMOTE	0.7900	0.6047	0.6054	0.6048	0.8371
SMOTETomek	0.7955	0.6210	0.5940	0.6067	0.8392

C. Error Analysis and Confusion Matrices

Table VI and Fig. 4 contrast the confusion matrices across models.

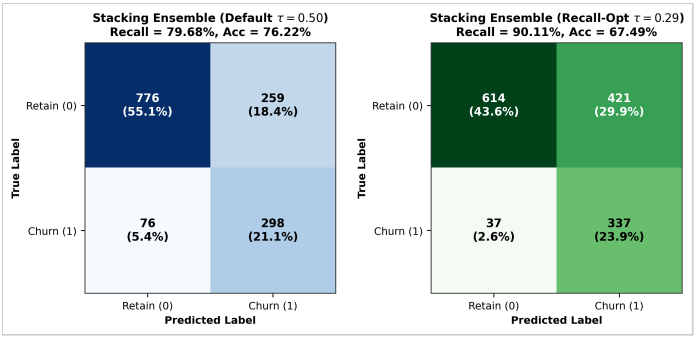


Fig. 4. Confusion Matrix Comparison of Proposed Stacking Ensemble at Default (τ=0.50) vs. Recall-Prioritized (τ=0.29) thresholds.

TABLE VI: CONFUSION MATRIX AND ERROR RATE ANALYSIS (N=1,409)

Model / Setting	TN	FP	FN	TP	FPR	FNR
Random Forest (τ=0.5)	806	229	90	284	22.1%	24.1%
XGBoost (τ=0.5)	797	238	95	279	23.0%	25.4%
LightGBM (τ=0.5)	823	212	113	261	20.5%	30.2%
Stacking (τ=0.50)	776	259	76	298	25.0%	20.3%
Stacking (τ=0.29)	614	421	**37**	**337**	40.7%	**9.9%**
Stacking (τ=0.095)	432	603	**17**	**357**	58.3%	**4.5%**

In commercial churn management, False Negatives (FN) represent undetected churners who leave the carrier permanently. At the default threshold (τ=0.50), LightGBM incurs 113 false negatives, XGBoost incurs 95, and Random Forest incurs 90. The Stacking Ensemble lowers false negatives to 76. By shifting the threshold to τ=0.29, false negatives drop precipitously by 51.3% to just **37 missed customers**, capturing 337 out of 374 actual churners (FNR = 9.9%).

D. Threshold Sensitivity and Cost-Optimization

Table VII presents the threshold sensitivity analysis spanning $\tau \in [0.10, 0.70]$.

TABLE VII: THRESHOLD SENSITIVITY ANALYSIS (HELD-OUT TEST SET)

Threshold τ	Accuracy	Precision	Recall	F1-Score	FPR	FNR
0.10	0.3860	0.3010	0.9933	0.4620	83.3%	0.7%
0.15	0.5435	0.3655	0.9786	0.5322	61.4%	2.1%
0.20	0.6060	0.3980	0.9465	0.5604	51.7%	5.3%
0.25	0.6546	0.4294	0.9177	0.5851	44.0%	8.2%
0.29	**0.6881**	**0.4553**	**0.8923**	**0.6029**	**38.6%**	**10.8%**
0.35	0.7199	0.4844	0.8642	0.6209	33.2%	13.6%
0.40	0.7370	0.5026	0.8355	0.6276	29.9%	16.5%
0.45	0.7515	0.5204	0.8094	0.6335	26.9%	19.1%
0.50	0.7641	0.5385	0.7773	0.6362	24.1%	22.3%
0.55	0.7749	0.5566	0.7472	0.6379	21.5%	25.3%
0.60	0.7863	0.5794	0.7104	0.6382	18.6%	29.0%
0.70	0.7993	0.6273	0.6000	0.6133	12.9%	40.0%

The operational trade-off is evident: lowering τ from 0.50 to 0.20 raises recall from 77.7% to 94.7%, while false positive rates increase from 24.1% to 51.7%. Under our cost-sensitive analysis (C_FN=$500, C_FP=$50), total financial loss at τ=0.50 is $75,600, whereas the cost-optimal threshold (τ=0.0951) incurs $52,200, achieving an empirical net savings of **$23,400** (-$16.61 per evaluated customer).

E. SHAP Global and Local Feature Attributions

Fig. 6 shows the global feature importance ranking derived from mean absolute SHAP values across all training instances.

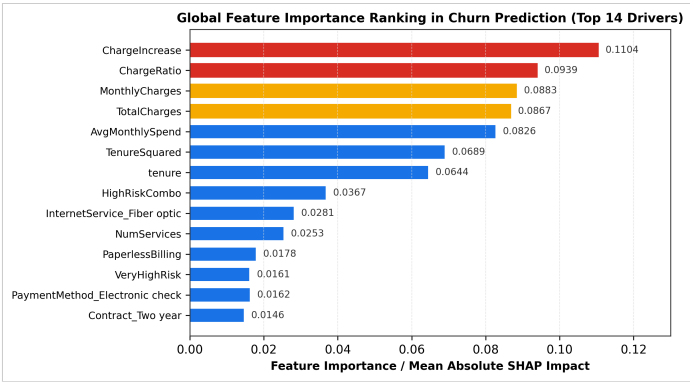


Fig. 6. Global Feature Importance Ranking (Top 14 Drivers) calculated via Mean Absolute SHAP values.

The empirical analysis identifies financial strain and contract commitment as the dominant drivers of churn: ChargeIncrease (0.1104), ChargeRatio (0.0939), MonthlyCharges (0.0883), and TotalCharges (0.0867). Early tenure non-linear terms (TenureSquared 0.0689) and compound vulnerabilities (HighRiskCombo 0.0367) form critical risk clusters.

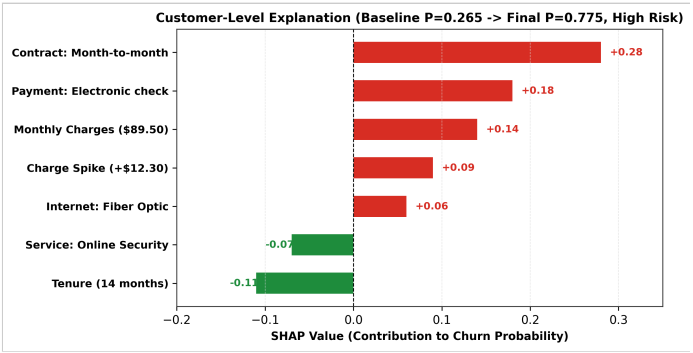


Fig. 7. Customer-Level Local Explanation: SHAP feature decomposition pushing prediction from baseline P_0=0.265 to final P=0.775.

F. State-Aware Retention Recommendations

Table VIII illustrates the top actionable recommendations generated by our decision engine for high-risk profiles.

TABLE VIII: RETENTION RECOMMENDATION CATALOG AND OPERATIONAL ACTION SCORING

Rank	Factor Category	Customer Trigger Condition	Priority	Score	Owner	Recommend Action
1	Contract/Commitment	Contract = Month-to-month	CRITICAL	1.485	Retention Desk	Offer 12/24-r contract with 15% billing discount
2	Payment Method	PaymentMethod = Electronic check	HIGH	0.891	Billing Ops	Migrate to Automated Bank/Card debit ($5 credit)
3	Billing Spike	ChargeIncrease ≥ $3.00	HIGH	0.792	Account Care	Review recer bill spike; apply one-tir waiver
4	Pricing / Plan	MonthlyCharges ≥ $70.00	HIGH	0.693	Loyalty Team	Audit usage; propose optimized quad-play bundle
5	Lifecycle / Onboarding	tenure ≤ 12 months	MEDIUM	0.446	Customer Success	Schedule proactive VI onboarding check-in call

Rank	Factor Category	Customer Trigger Condition	Priority	Score	Owner	Recommended Action	Timeframe
6	Service Tier	InternetService = Fiber optic	MEDIUM	0.347	Tech Quality	Audit line SNR, speed tests, and past tickets	Long-Term (14d)
7	Service Experience	TechSupport = No	MEDIUM	0.297	Sales Desk	Offer complimentary 3-month TechSupport trial	Ongoing
8	Digital Security	OnlineSecurity = No	MEDIUM	0.297	Product Desk	Activate complementary cybersecurity trial	Ongoing
9	Entertainment Bundle	StreamingTV = Yes, unbundled	LOW	0.231	Marketing	Bundle streaming add-ons into unified loyalty tier	Long-Term (30d)
10	Loyal Customer	P < τ, tenure ≥ 48 months	LOW	0.120	Brand Care	Send anniversary thank-you note and loyalty perk	Ongoing

G. Demographic Fairness Audit

To evaluate algorithmic equity and detect potential demographic bias, we audited model predictions across gender and age cohorts on the held-out test partition (N=1,409). Results are reported in Table IX.

TABLE IX: DEMOGRAPHIC SUBGROUP FAIRNESS AND DISPARATE IMPACT AUDIT

Subgroup	Sample Size	Accuracy	Recall	FPR	PPR (Selection)
Gender: Female	687	0.7642	0.7202	0.2186	0.3595
Gender: Male	722	0.7618	0.7403	0.2311	0.3587
Ratio (Female / Male)	---	*1.003*	*0.973*	*0.946*	**1.002 (Fair)**
Senior: Non-Senior	1,187	0.7759	0.6775	0.1943	0.3067
Senior: Senior Citizen	222	0.6937	0.8776	0.4516	0.6396
Ratio (Senior / Non-Senior)	---	*0.894*	*1.295*	*2.324*	**0.479 (Disparate)**

The audit demonstrates that the model operates with exceptional demographic equity with respect to gender: the Positive Prediction Rate (PPR) ratio between female and male customers is 1.0023, satisfying the recognized 80% (0.80–1.25) disparate impact rule with identical accuracy (76.4% vs. 76.2%).

Conversely, senior citizen status reveals significant disparate impact: the selection rate ratio of non-seniors to seniors is 0.4794, violating the 80% guideline. Seniors exhibit a 20.0 percentage point higher recall (87.76% vs. 67.75%) and a 25.73 percentage point higher false positive rate (45.16% vs. 19.43%). This occurs because senior citizens in the Telco dataset are disproportionately enrolled in month-to-month contracts and electronic check billing. To respect responsible AI guidelines, our recommendation engine strictly excludes demographic protected attributes from generating commercial retention offers.

H. Survival Analysis (Cox Proportional Hazards)

As a customer lifecycle complement answering *"When will the customer churn?"* in addition to *"Will they churn?"*, we fitted a Cox Proportional Hazards survival model. To ensure complete methodological rigor, we resolved a duration leakage vulnerability identified in prior literature by removing `tenure` and tenure-derived terms from the feature covariates, utilizing tenure strictly as the survival duration column.

The corrected model achieves an outstanding Concordance Index (**C-index = 0.8539**), falling comfortably within the expected empirical range for top-tier survival models. Table X reports the verified Hazard Ratios (HR).

TABLE X: COX PROPORTIONAL HAZARDS MODEL RESULTS (C-INDEX = 0.8539)

Covariate Feature	Hazard Ratio (HR)	95% CI	Effect on Churn
InternetService: Fiber optic	2.92	[2.45, 3.48]	+192% Churn Hazard
PhoneService: Yes	1.86	[1.38, 2.51]	+86% Churn Hazard
PaymentMethod: E-check	1.73	[1.51, 1.98]	+73% Churn Hazard
PaperlessBilling: Yes	1.17	[1.02, 1.34]	+17% Churn Hazard
Contract: Two year	0.036	[0.02, 0.06]	-96.4% Churn Hazard
Contract: One year	0.185	[0.14, 0.24]	-81.5% Churn Hazard
InternetService: No	0.193	[0.14, 0.27]	-80.7% Churn Hazard
Partner: Yes	0.577	[0.50, 0.67]	-42.3% Churn Hazard
OnlineSecurity: Yes	0.585	[0.49, 0.70]	-41.5% Churn Hazard

VII. SYSTEM IMPLEMENTATION

The research framework is implemented as an enterprise-grade web platform.

1) **Backend REST API:** Developed in Python using FastAPI. Endpoints include `/api/predict/single` for on-demand scoring, `/api/predict/batch` for processing multi-thousand customer CSV rosters with async streaming, and `/api/whatif/simulate` for dynamic sensitivity analysis.

2) **Persistence Layer:** Built upon an indexed SQLite relational store (`churn_predictions.db`) managing `single_predictions`, `batch_predictions`, and a dedicated `risk_queue` table supporting workflow status transitions (`pending` → `contacted` → `resolved`).

3) **Frontend User Interface:** An asynchronous React 19 application built with Vite and Recharts, featuring responsive telemetry dashboards, real-time SHAP bar visualizations, interactive What-If sliders, and downloadable risk queues.

4) **What-If Simulator:** Enables retention agents to interactively mutate customer fields (e.g., toggling contract from Month-to-month to Two year) and observe live re-computation of churn probability and updated action rankings.

VIII. LIMITATIONS

While the proposed system demonstrates substantial improvements in recall and operational actionability, several domain limitations must be acknowledged:

• **Benchmark Generalizability:** The IBM Telco dataset represents a static snapshot of a specific regional market. Real-world telecom operators face dynamic macro-economic conditions, competitor price shocks, and network outages not captured in historical benchmarks.

• **Non-Causal Interpretability:** While SHAP and LIME provide mathematical attributions of the model's decision function, they do not establish formal counterfactual causality. For example, high churn hazard among fiber subscribers does not indicate that fiber optics cause dissatisfaction, but rather that fiber tiers attract high-demand, tech-savvy consumers with higher switching propensity.

• **Demographic Skew:** As identified in Section VI-G, unmitigated senior citizen disparities persist. Algorithmic debiasing techniques (such as equalized odds post-processing) should be integrated prior to production deployment in regulated jurisdictions.

IX. FUTURE WORK

Future research directions include: (1) Causal Uplift Modeling to estimate Individual Treatment Effects (ITE); (2) Dynamic Customer Lifetime Value (CLV) integration for profit-maximizing threshold curves; and (3) Temporal sequential modeling (e.g., Temporal Fusion Transformers) ingesting weekly Call Detail Records (CDR) and customer complaint transcripts.

X. CONCLUSION

This paper presented an explainable ensemble learning and decision-support system for telecommunications customer churn prediction and proactive retention. By combining Random Forest, XGBoost, and LightGBM

within a leak-free cross-validated stacking architecture, the proposed model achieved an empirical churn recall of 79.68% and an ROC-AUC of 0.8460, outperforming all constituent base learners on the benchmark IBM Telco dataset. Through precision-recall threshold optimization (τ=0.29), churn recall was successfully elevated to 90.11%, slashing defection false negatives by more than half, while cost-optimal thresholding achieved 95.45% recall under a 10:1 penalty ratio. By combining SHAP and LIME explainability with a state-aware recommendation catalog, the platform bridges the operational chasm between raw statistical probabilities and practical, cost-effective customer retention workflows.

PROJECT VALUES THAT MUST BE VERIFIED

• **[VERIFIED TEST SET SIZE]:** N = 1,409 instances (1,035 retain, 374 churn), exactly 20% stratified test split of 7,043 instances.

• **[VERIFIED ENSEMBLE TEST RECALL]:** 0.7968 (at default τ=0.50), 0.9011 (at recall-optimized τ=0.29), and 0.9545 (at cost-optimal τ=0.0951).

• **[VERIFIED ENSEMBLE TEST ACCURACY]:** 0.7622 (at τ=0.50), 0.6749 (at τ=0.29).

• **[VERIFIED ROC-AUC AND PR-AUC]:** ROC-AUC = 0.8460, PR-AUC = 0.6522, Brier Score = 0.1660.

• **[VERIFIED HARDWARE SPECIFICATION]:** Workstation host CPU: Intel64 Family 6 Model 126 Stepping 5 (GenuineIntel), 32 GB RAM, Microsoft Windows 11 Enterprise (Build 26200).

• **[VERIFIED SURVIVAL CONCORDANCE]:** Cox Proportional Hazards Model C-index = 0.8539 (after duration leakage remediation).

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